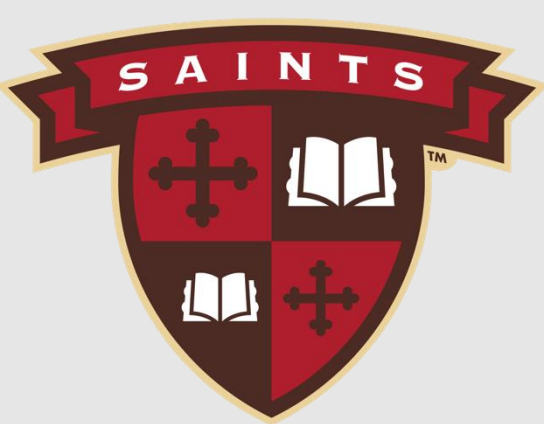


Optimizing Combat in Dungeons & Dragons: A Unity-Based Simulation



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Background

- During Dungeons & Dragons (D&D) battles, players make quick choices, but it's often hard to predict if those decisions will lead to success when preparing the game ahead of time, which can mess with pacing or enjoyment.
- Predicting how a team will perform in battle ahead of time is tricky for those setting up a story, and DMs(Dungeon Masters) often rely on guesswork, which can lead to unbalanced or unpredictable encounters.
- This project introduces a combat simulator that allows users to create teams, assign strategies, and run test battles based on a random initiative. It gives users or DMs a clearer sense of how different team compositions and tactics might play out, without needing to roll any dice.



Objectives

- Create tokens that can be assigned to user-defined teams and positioned within a virtual battlefield. A user-friendly interface was designed to allow players to customize each token's key attributes, making it easier to test a wide variety of tactical scenarios based on different character builds.
- Research Dungeons & Dragons character sheets to identify the most relevant attributes for simulation, using that information to help design a clear UML class structure and build editable token features that reflect those stats.
- Allow for the selection of various attack strategies, such as prioritizing the lowest-health target or using randomized targeting.
- Create a randomized initiative order that allows users to run multiple iterations that help assess survival probabilities and evaluate the consistency of different strategies.

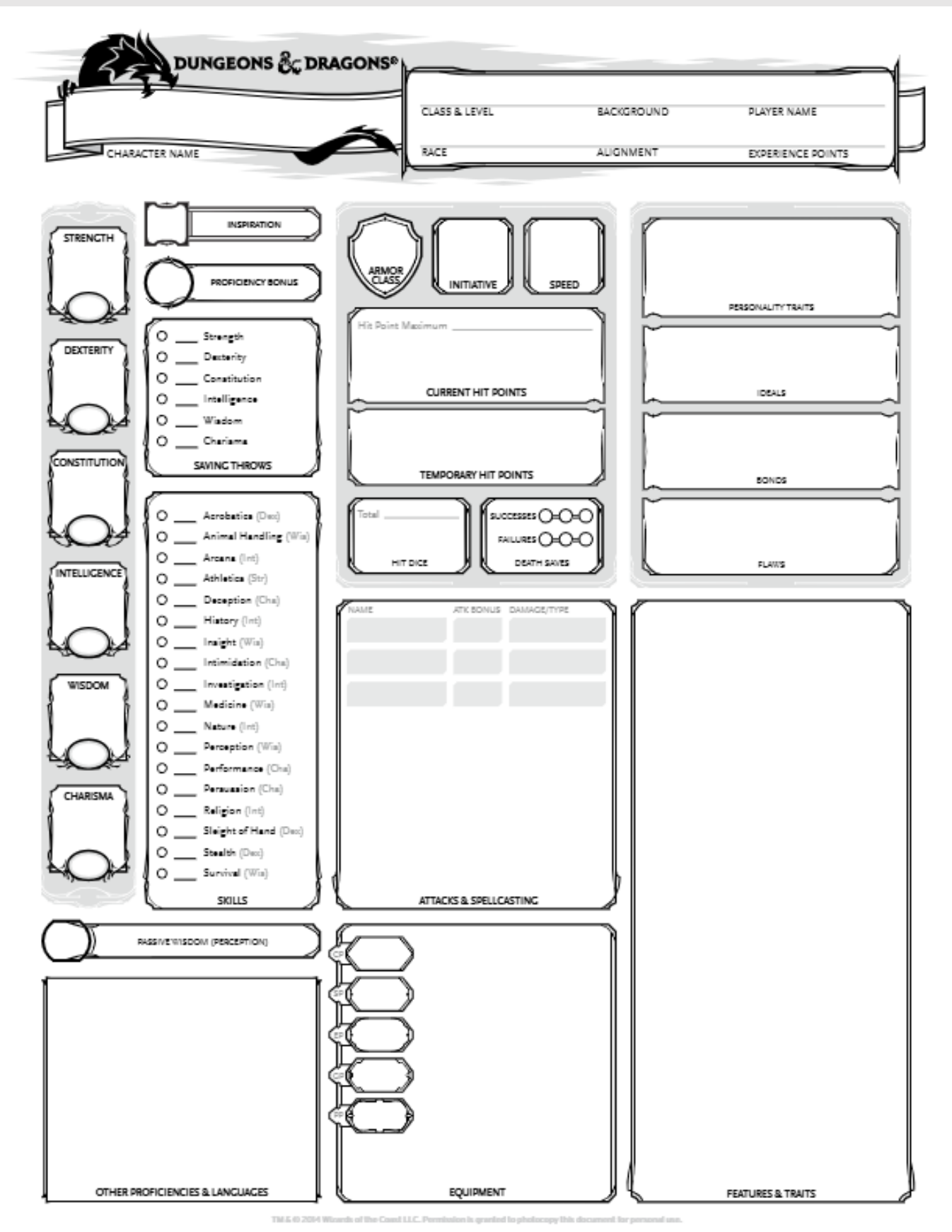


Figure 1.
In this photo you can see a character sheet used in D&D, a place to collect needed facts about a given character.

Methodology

- I used object-oriented principles like class inheritance to build a hierarchy, making it easier to implement future features while also keeping the structure more organized and easier to understand when working with many interconnected scripts.
- Used Unity's UI system to design an interactive interface. Users can create and customize tokens, assign them to teams, and modify attributes like health, armor class, speed, and the ability scores.
- Wrote and implemented a turn-based combat system that includes initiative rolls and changeable targeting logic, and customizable strategies.

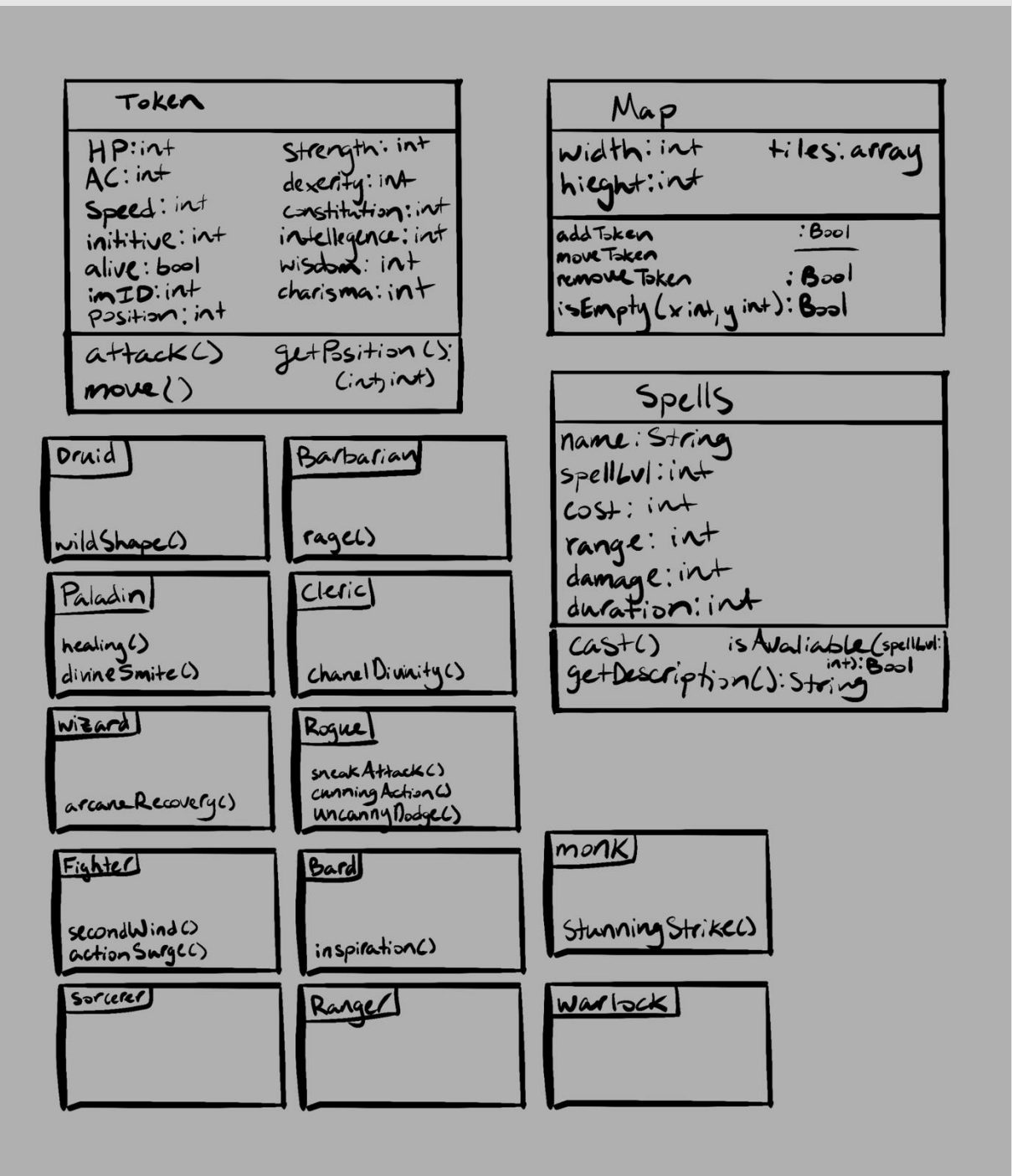


Figure 2.
In this photo you can see the outline of my UML(Unified Modeling Language) for the classes.

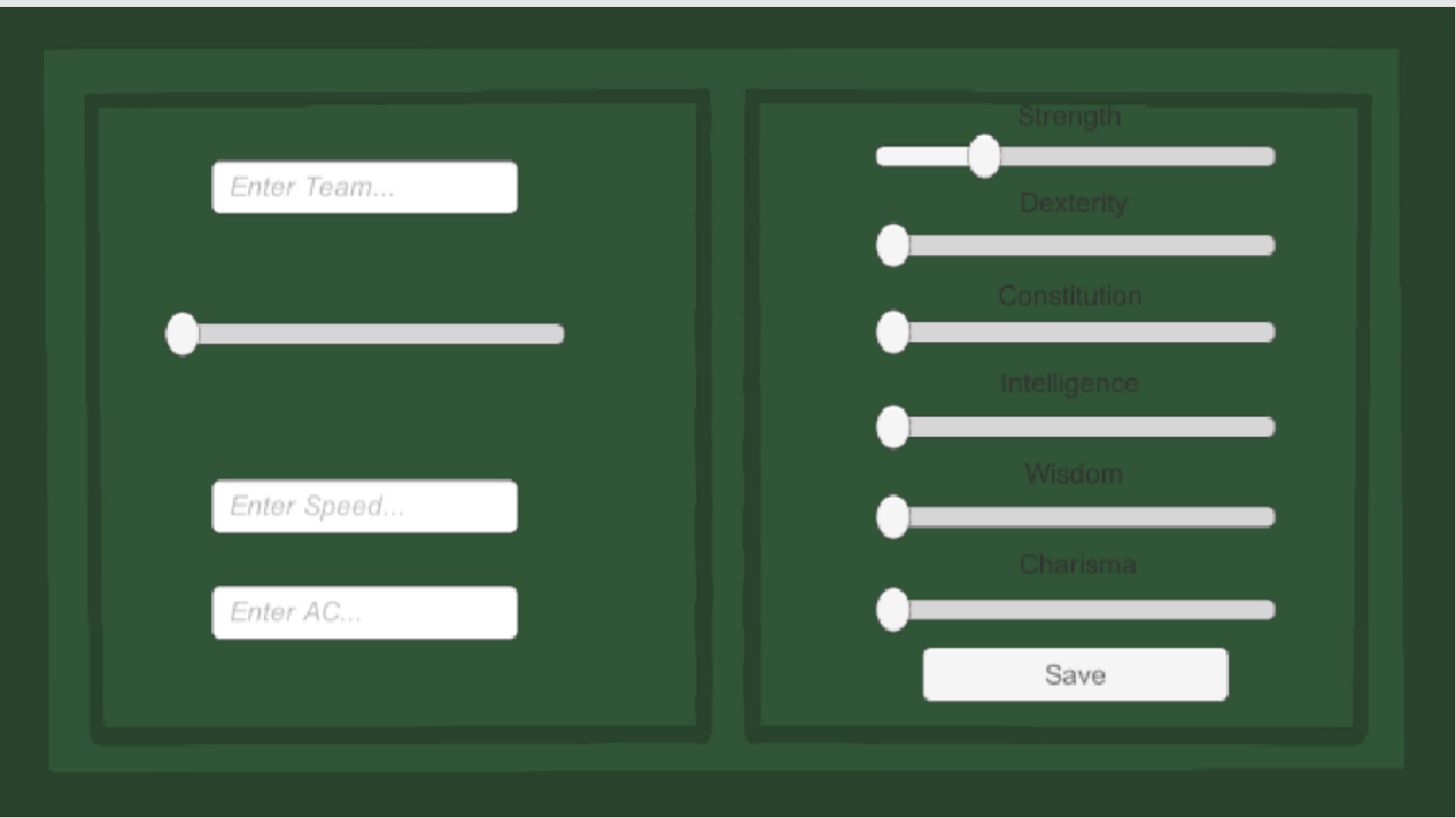


Figure 3.
In this photo you can see the set up of token menu, which is where you can change attributes to be personalized to each token.

Simulator

This simulator serves as a functional tool for visualizing and analyzing D&D combat encounters. Built with flexibility in mind, it allows users to create tokens, assign them to teams, and select from multiple targeting strategies to simulate combat scenarios. The system is operational and ready for users to explore tactical interactions and team dynamics on their own. It offers a practical way to engage with the complexities of combat outside of live gameplay as well as opens an opportunity for potential testing.

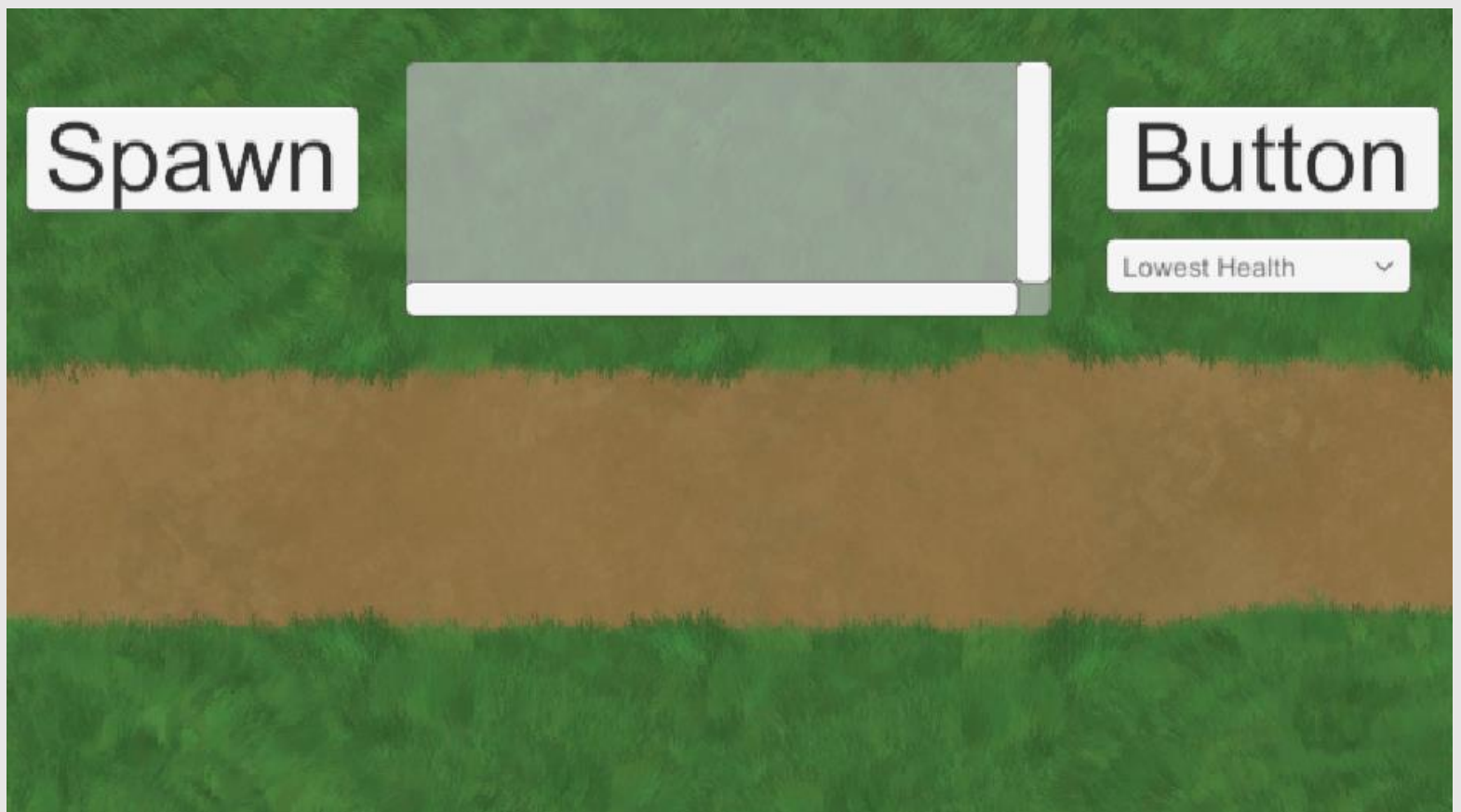


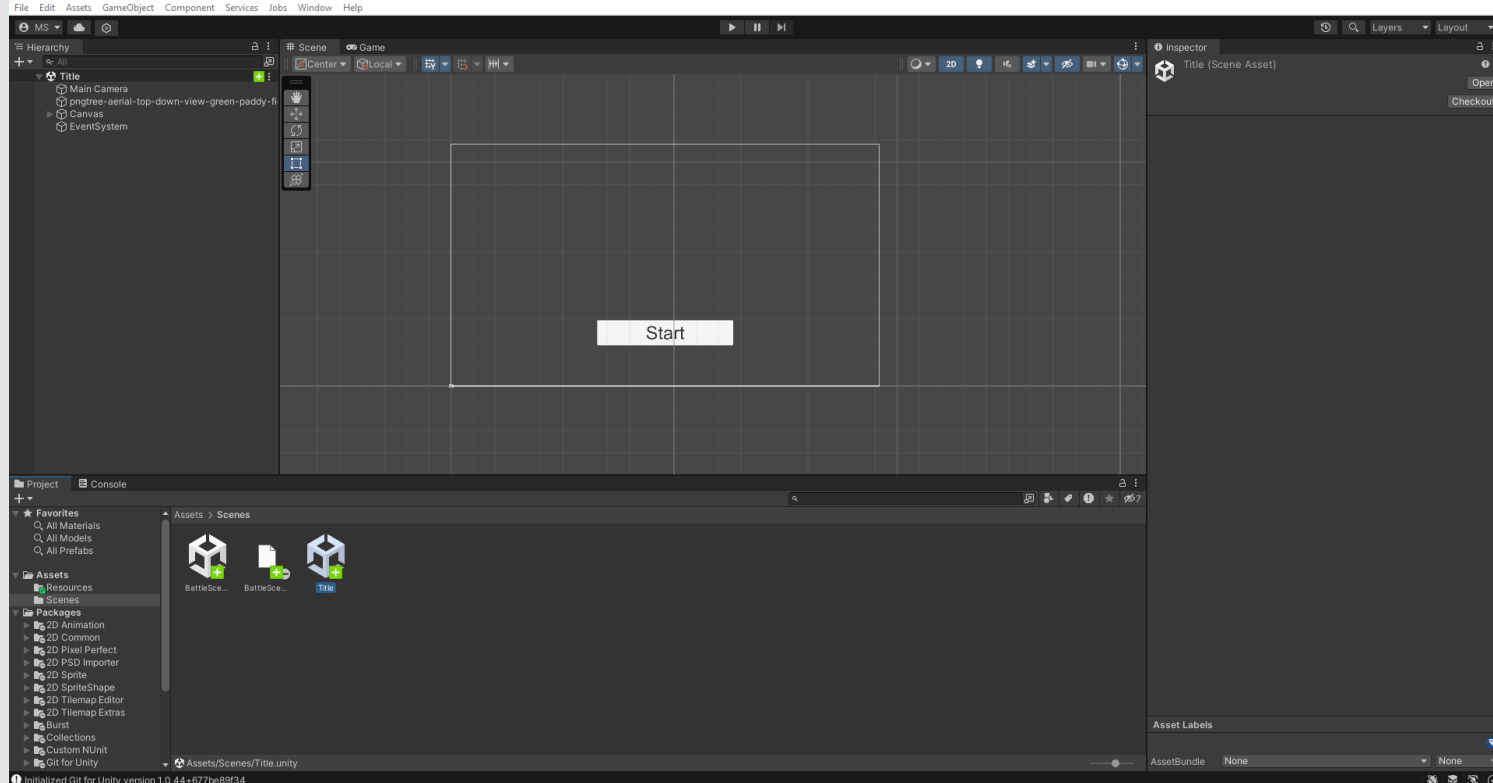
Figure 4.
In this photo you can see the main screen of the simulator, the battle scene. This includes a button to spawn in more tokens, a history text box, start button, and a drop-down menu to select different strategies.

Tools

- Unity**
Used as the main engine to build and run the simulator. It handled everything from placing tokens on the field to keeping track of battle logic.
- Unity Version Control**
Helped me keep track of changes throughout development, making it easier to revisit earlier versions and collaborate.
- Visual Studio Code**
My main coding environment for writing and editing C# scripts. It kept things clean and efficient while working through logic and functionality.
- C#**
The programming language I used to build every aspect of the simulator's functionality with the use of scripts.
- Procreate**
Used not only for sketching early UI concepts but also to design and create original backgrounds and token artwork, adding a personalized and creative visual layer to the project.



Figure 5.
In this photo you can see the layout and look of the game engine, Unity.



Future Work

In the future, I aim to explore a wider range of strategies that players commonly use in D&D, which could make the simulator even more useful for players and DMs who want to better understand the flow of combat. I also plan to fully implement features like character classes and associated abilities, items, and spells. These additions would introduce more depth and allow the simulator to handle complex and less predictable combat scenarios that are often shaped by those mechanics. I'd also like to add more token designs that align with each specific class added and look into redesigning buttons and text.

Acknowledgments

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